

2_Review_&_Analyze_Requirements → Requirement_Analysis:

Requirement Analysis Document

Project Title

Data-Driven Testing of SauceDemo Login Functionality Using Selenium WebDriver, TestNG, and CSV File

1. Introduction

Requirement Analysis is the first and one of the most important phases of the Software Testing Life Cycle (STLC). In this phase, testers study and understand the business requirements, functional requirements, and user expectations of the application before designing test cases.

The purpose of Requirement Analysis is to identify what needs to be tested, determine the testing scope, understand system behavior, identify risks, and ensure complete test coverage.

For this project, the Login functionality of the SauceDemo application is analyzed to understand how users authenticate themselves and gain access to the application.

2. Requirement Gathering

The requirements for this project are gathered from the SauceDemo Login page and application behavior.

Application URL:

<https://www.saucedemo.com>

The Login page contains:

- Username Input Field
- Password Input Field
- Login Button

- Error Message Section

The application allows registered users to log in using valid credentials and access the Products page.

3. Business Requirement Analysis

Business requirements describe what the application should achieve from the user's perspective.

BR-001

The system shall allow registered users to log in using valid credentials.

BR-002

The system shall prevent unauthorized users from accessing the application.

BR-003

The system shall display an appropriate error message when invalid credentials are entered.

BR-004

The system shall validate mandatory fields before processing login requests.

BR-005

The system shall redirect authenticated users to the Products page after successful login.

4. Functional Requirements

Functional requirements describe the specific functionality that the system must perform.

FR-001

The Username field should accept valid usernames.

FR-002

The Password field should accept valid passwords.

FR-003

The Login button should authenticate users based on entered credentials.

FR-004

The application should allow successful login when correct credentials are entered.

FR-005

The application should reject invalid credentials.

FR-006

The application should display an error message for invalid login attempts.

FR-007

The application should display an error message when Username is left blank.

FR-008

The application should display an error message when Password is left blank.

FR-009

The application should display an error message when both fields are blank.

FR-010

The application should redirect users to the Products page after successful authentication.

5. Non-Functional Requirements

Non-functional requirements define quality attributes of the application.

NFR-001 – Usability

The Login page should be user-friendly and easy to navigate.

NFR-002 – Performance

The login process should complete within an acceptable response time.

NFR-003 – Reliability

The login functionality should work consistently without failures.

NFR-004 – Compatibility

The application should function correctly on supported web browsers.

NFR-005 – Availability

The login service should be available whenever users access the application.

6. Identified Functional Components

The following components are identified during requirement analysis:

Component 1: Username Field

Purpose:

Allows users to enter their username.

Validation Requirements:

- Should accept valid usernames.
 - Should not allow blank values during login.
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Component 2: Password Field

Purpose:

Allows users to enter their password.

Validation Requirements:

- Should accept valid passwords.
 - Should not allow blank values during login.
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Component 3: Login Button

Purpose:

Initiates the login process.

Validation Requirements:

- Should validate entered credentials.
 - Should redirect valid users.
 - Should display errors for invalid users.
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Component 4: Error Message Section

Purpose:

Displays validation and authentication errors.

Validation Requirements:

- Proper message should appear when login fails.
 - Error message should be visible to users.
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7. Positive Test Conditions

The following positive conditions are identified:

PC-001

Login using valid username and valid password.

Expected Result:

User should be redirected to Products page.

PC-002

Login using performance_glitch_user credentials.

Expected Result:

User should successfully log in.

PC-003

Login using problem_user credentials.

Expected Result:

User should successfully log in.

8. Negative Test Conditions

The following negative conditions are identified:

NC-001

Login using invalid username.

Expected Result:

Error message should be displayed.

NC-002

Login using invalid password.

Expected Result:

Error message should be displayed.

NC-003

Login using invalid username and invalid password.

Expected Result:

Error message should be displayed.

NC-004

Login with blank username.

Expected Result:

Username required error should be displayed.

NC-005

Login with blank password.

Expected Result:

Password required error should be displayed.

NC-006

Login with both fields blank.

Expected Result:

Mandatory field validation should be displayed.

NC-007

Login using locked_out_user.

Expected Result:

User should not be allowed to log in.

Appropriate error message should be displayed.

9. Assumptions

The following assumptions are considered:

- The application URL is accessible.
- Test users are available.
- Internet connectivity is stable.

- Browser is functioning properly.
 - Test environment is available during execution.
 - Login functionality is active and operational.
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10. Constraints

The following constraints are identified:

- Testing is limited to the Login Module.
 - Performance testing is not included.
 - Security testing is not included.
 - Mobile testing is not included.
 - Testing is performed only on supported browsers.
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11. Risk Analysis

Potential risks identified during requirement analysis are:

Risk 1

Application downtime during testing.

Impact:

Test execution may be delayed.

Mitigation:

Retest once the application becomes available.

Risk 2

Requirement changes.

Impact:

Test cases may need modification.

Mitigation:

Maintain traceability and update documents.

Risk 3

Invalid test data.

Impact:

Incorrect test results.

Mitigation:

Validate test data before execution.

Risk 4

Browser compatibility issues.

Impact:

Unexpected application behavior.

Mitigation:

Execute tests on supported browsers.

12. Requirement Understanding Summary

After analyzing the Login module requirements, the following understanding was achieved:

- Users must provide valid credentials to access the application.
- Authentication should occur upon clicking the Login button.
- Invalid credentials should be rejected.
- Mandatory field validation should be enforced.

- Error messages should be displayed appropriately.
- Successful authentication should redirect users to the Products page.

The identified requirements provide sufficient information to proceed with Test Planning, Test Scenario Creation, Test Case Design, Manual Test Execution, and Automation Test Development.
